Computer Vision: Machine Learning based Facial Recognition, Autonomous Driving, and Image Processing

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Date: 2026/04/12

考试概览

- 期中考试形式

  --10道多选题

  --重点考察概念理解，不考复杂推导

- 考题分布：

  --分割 (Seg)　— 5-6题

  --机器学习 (ML) — 1-2题

  --检测 (Det)　— 1-2题

  --配准 (Reg)　— 1-2题

第一部分：Images 基础

- **颜色空间：** RGB（线性） vs HSV（非线性，分离亮度和颜色）
- **高斯滤波 vs 中值滤波：**

  高斯 — 线性，平滑但模糊边缘

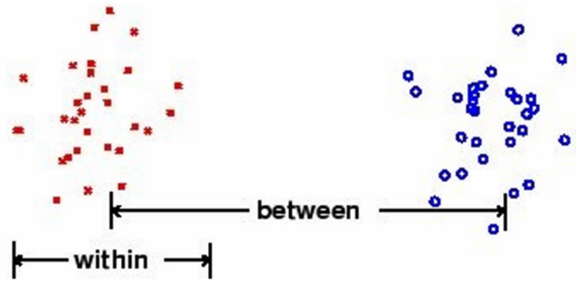
  中值 — 非线性，去椒盐噪声且保边缘
- **傅里叶变换：** 空间域卷积 = 频率域乘法 (Convolution Theorem)
- **图像金字塔：** Gaussian pyramid → 后面SIFT会用到
- **纹理描述：** GLCM → Contrast / Homogeneity / Entropy / Energy

The Otsu method

The optimal threshold minimizes the intra-class variance:

$$t^* = \min_{1 \leq t \leq L} \{\sigma_W^2(t) | \sigma_W^2 = w_0(t)\sigma_0(t)^2 + w_1(t)\sigma_1^2(t)\}$$

It can be shown that this is equivalent to maximising the between class variance.



K-means clustering

Step 1 - Initialization:

Randomly select k data points as cluster centers

Determine $\mathbf{S}^0, \mu_i^t$

Step 2 - Assignment Step:

Assign each observation to the closest cluster mean

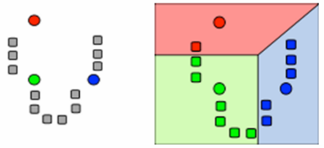
$$\min_i ||x - \mu_i^t||$$

Determine $\mathbf{S}^t$

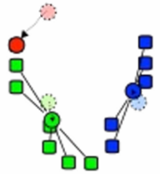
Step 3 - Update Step

Update the cluster means μ_i^{t+1}

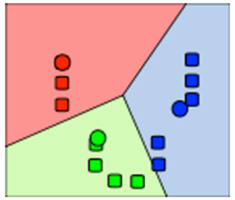
Interate Step 2 and Step 3 until convergence



Initialization



Assignment



Update

K-means initialisation

- Random initialisation
- Explicit labelling of a subset
- K-means++ initialisation method:
  - Choose one center uniformly at random from among the data points.
  - For each data point *x*, compute D(*x*), the distance between *x* and the nearest center that has already been chosen.
  - Choose one new data point at random as a new center, using a weighted probability distribution where a point *x* is chosen with probability proportional to $D(x)^2$.
  - Repeat Steps 2 and 3 until *k* centers have been chosen.
  - Now that the initial centers have been chosen, proceed using standard *k*-means clustering.

第二部分：Segmentation A

Simple linear iterative clustering (SLIC)

Algorithm 1 SLIC superpixel segmentation

/* *Initialization* */
Initialize cluster centers $C_k = [l_k, a_k, b_k, x_k, y_k]^T$ by sampling pixels at regular grid steps S.
Move cluster centers to the lowest gradient position in a 3×3 neighborhood.
Set label $l(i) = -1$ for each pixel i.
Set distance $d(i) = \infty$ for each pixel i.
repeat
  /* *Assignment* */
  for each cluster center C_k **do**
    for each pixel i in a $2S \times 2S$ region around C_k **do**
      Compute the distance D between C_k and i.
      if $D < d(i)$ **then**
        set $d(i) = D$
        set $l(i) = k$
      end if
    end for
  end for
  /* *Update* */
  Compute new cluster centers.
  Compute residual error E.
until $E \leq$ threshold

SLIC – examples

Fig. 1: Images segmented using SLIC into superpixels of size 64, 256, and 1024 pixels (approximately).

其他分割方法（了解）

- Watershed：梯度图→地形图，从标记点"注水"扩展
- 形态学操作：

    Opening = 先腐蚀后膨胀 → 去小噪点

    Closing = 先膨胀后腐蚀 → 填小空洞
- Connected Component Labelling：基于像素连通性分组
- Mean Shift：非参数密度估计，沿梯度方向移动，无需指定K

Affinity measures

We can measure the affinity of pixels, superpixels or segments

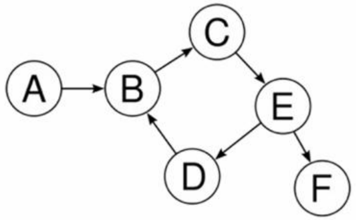
Affinity by distance:

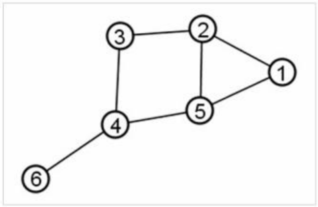
$$\mathrm{aff}(x,y) = exp\{-((x-y)^t(x-y)/2\sigma_d^2\}$$

Affinity by intensity:

$$\mathrm{aff}(x,y) = exp\{-((I(x)-I(y))^t(I(x)-I(y))/2\sigma_I^2\}$$

Pairwise affinities define a graph





- A graph is a set of vertices V and edge E that connects various pairs of vertices. Each edge can be represented as a pair of vertices.
- A directed graph is one in which edges (a,b) and (b,a) are distinct.
- An undirected graph is one in which no distinction is drawn between edges (a,b) and (b,a).

第三部分：Segmentation B

Normalized cuts

An alternative approach is to cut the graph V into connected components such that the cut is a small fraction of the total affinity within each group.

Cutting groups A and B

$$\mathrm{cut}(A,B) = \sum_{v \in A} \sum_{w \in B} \mathrm{aff}(v \to w)$$

Association with group A

$$\mathrm{assoc}(A,V) = \sum_{v \in V} \sum_{w \in A} \mathrm{aff}(v \to w)$$

J. Shi and J. Malik. Normalized cuts and image segmentation. *IEEE Trans. Pattern Anal. and Machine Intell.*, 22(8):888–905, 2000.

Interactive graph cuts

Labelling problems can be formulated in terms of an energy minimization:

$$E(A) = \underbrace{\sum_{p \in P} D_p(A)}_{\text{Penalty for Pixel labelling}} + \lambda \underbrace{\sum_{p,q} V_{p,q}(A)}_{\text{Smoothing term pixel interactions}} \quad \text{with} \quad \lambda \geq 0$$

Energy function

As we are interested in estimating the optimal boundary the energy term becomes:

$$E(A) = \underbrace{\sum_{p \in P} R_p(A)}_{\text{Regional term}} + \lambda \underbrace{\sum_{p,q} B_{p,q}\delta(A)}_{\text{Boundary term}} \quad \text{with} \quad \lambda \geq 0$$

Region metrics

Dice Similarity Index (DSC or SI):

- Ratio of number of pixels in the intersection of two volumes, to the mean volume (this is sometimes called the “mean overlap”, MO)

$$DSC = \frac{2(A \cap B)}{A + B}$$

Tanimoto coefficient (TC) or Jacquard index:

- Also called the “union overlap”

$$TC = \frac{A \cap B}{A \cup B}$$

Measures are related:

- Same at extremes (0,1) DSC>TC in between (hence DSC often favoured...!)

$$DSC = \frac{2TC}{TC + 1}$$

Validation of segmentation: Hausdorff Distance

Measures worst match between 2 curves

- Given two curves, X={x1,..,xn}, Y={y1,..ym}, calculate closest distance from each x_i to Y as:

$$d(\mathbf{x}_i, \mathbf{Y}) = \min_{j=1}^{m} \|\mathbf{y}_j - \mathbf{x}_i\|$$

- The Hausdorff distance is defined as the maximum of all minimum distances of x_i to Y_i and hence measures the disparity of X and Y:

$$HD = \max\left(\{d(\mathbf{x}_i, \mathbf{Y})\}\right)$$

- A symmetric Hausdorff metric exists, which also measures the closest distances from y_i to X, and the returns the maximum distance

$$HD_{symm}(X,Y) = \max\left(\max\left(\{d(\mathbf{x}_i, \mathbf{Y})\}\right) \max\left(\{d(\mathbf{y}_j, \mathbf{X})\}\right)\right)$$

Canny edge detector – core steps

- Filter image with derivatives of Gaussian
- Compute gradient magnitude and orientation
- Non-maximal suppression:
  - Thin multi-pixel wide "ridges" down to a single pixel width
- Thresholding and linking (hysteresis)
  - Define two thresholds (maxVal and minVal)
  - Remove weak edges and link broken edge segments based on the thresholds

1) Keep strong edges

$$|\nabla I(x)| > \mathrm{maxVal} \quad \text{for all} \quad x \in E_1$$

2) Remove weak edges

$$|\nabla I(x)| < \mathrm{maxVal} \quad \text{for all} \quad x \in E_2$$

3) Link edge segments

$$|\nabla I(x)| > \mathrm{minVal} \quad \text{for all} \quad x \in E_3$$

第四部分：Features

- SIFT 完整4步：

--第一步Scale-space Extrema Detection——在DoG金字塔找极值点，和26个邻居比较。

--第二步Keypoint Localization——Taylor展开精确定位，剔除低对比度点和边缘点。

--第三步Orientation Assignment——给关键点算主方向，实现旋转不变性。

--第四步Descriptor Formation——4×4子区域×8-bin方向直方图→128维。

- Blob Detection & Hessian：

-- Blob Detection — LoG (Laplacian of Gaussian)

尺度与blob大小匹配时响应最大

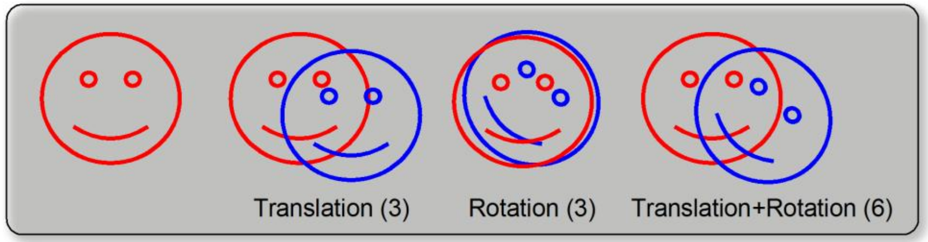
-- Hessian 矩阵

特征值分析 → 管状(一大一小) / 球状(两大) / 平坦(两小)

多尺度分析：检测不同大小的结构（Frangi滤波器用于血管）

第五部分：Registration

Global transformation models – Euclidian similarity



$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \qquad \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -h \\ 0 & 1 & -k \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

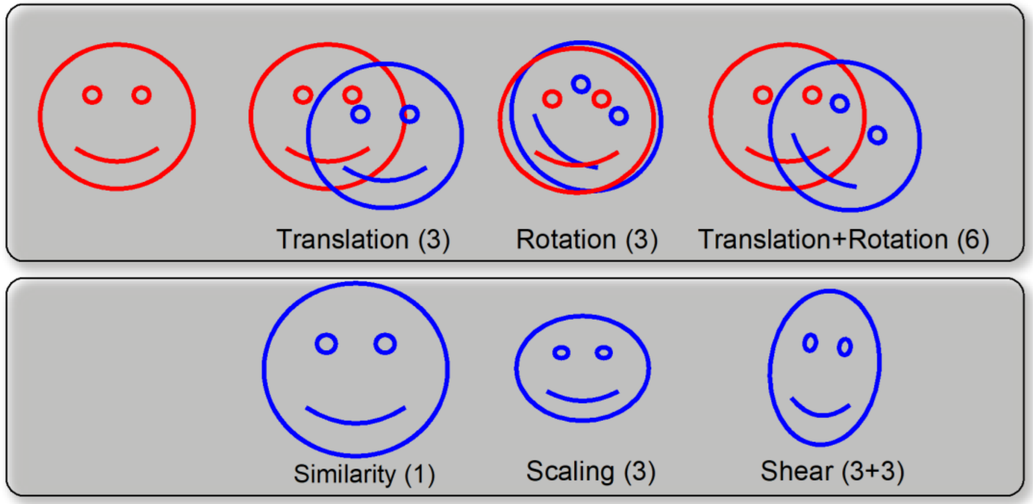
Translation in the plane

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos a & -\sin a & h \\ \sin a & \cos a & k \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \cos a & \sin a & -h\cos a - k\sin a \\ -\sin a & \cos a & h\sin a - k\cos a \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

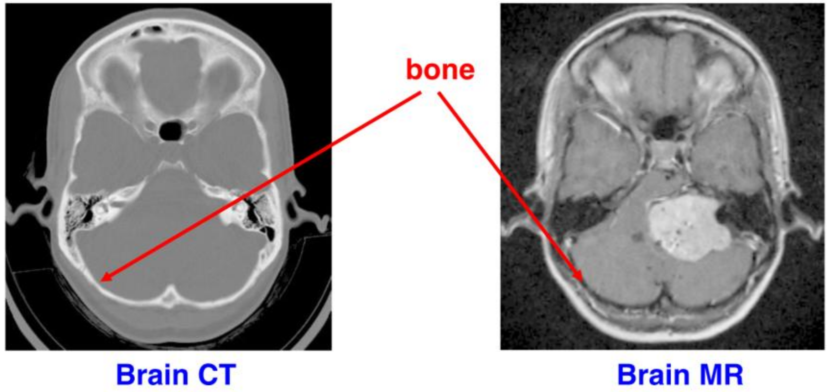
Rotation in the plane

Global transformation models – affine transform



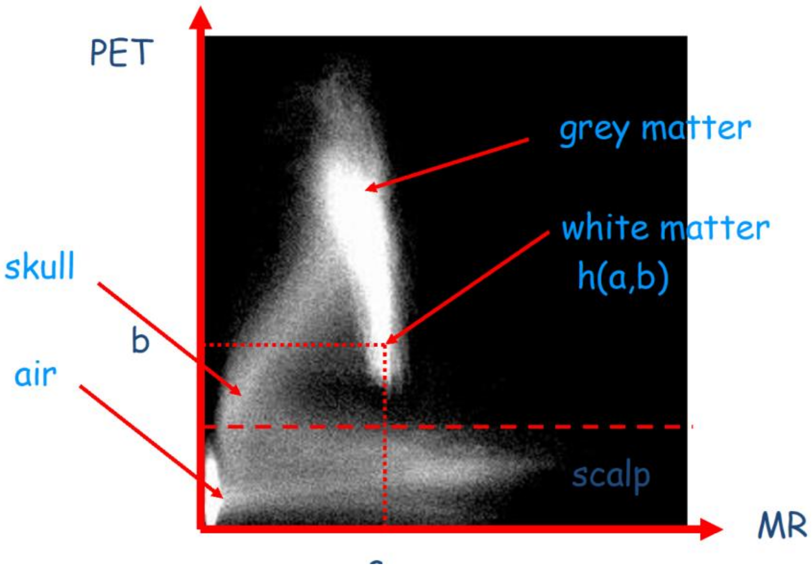
第五部分：Registration

Problem: extension to multiple modalities



How do we match structures which have different image appearance in different image modalities as in the case matching intensities by least squares (say) will not work?

2D Joint Intensity Histograms



A joint intensity histogram for the same object (a brain slice) imaged with MRI and with PET.

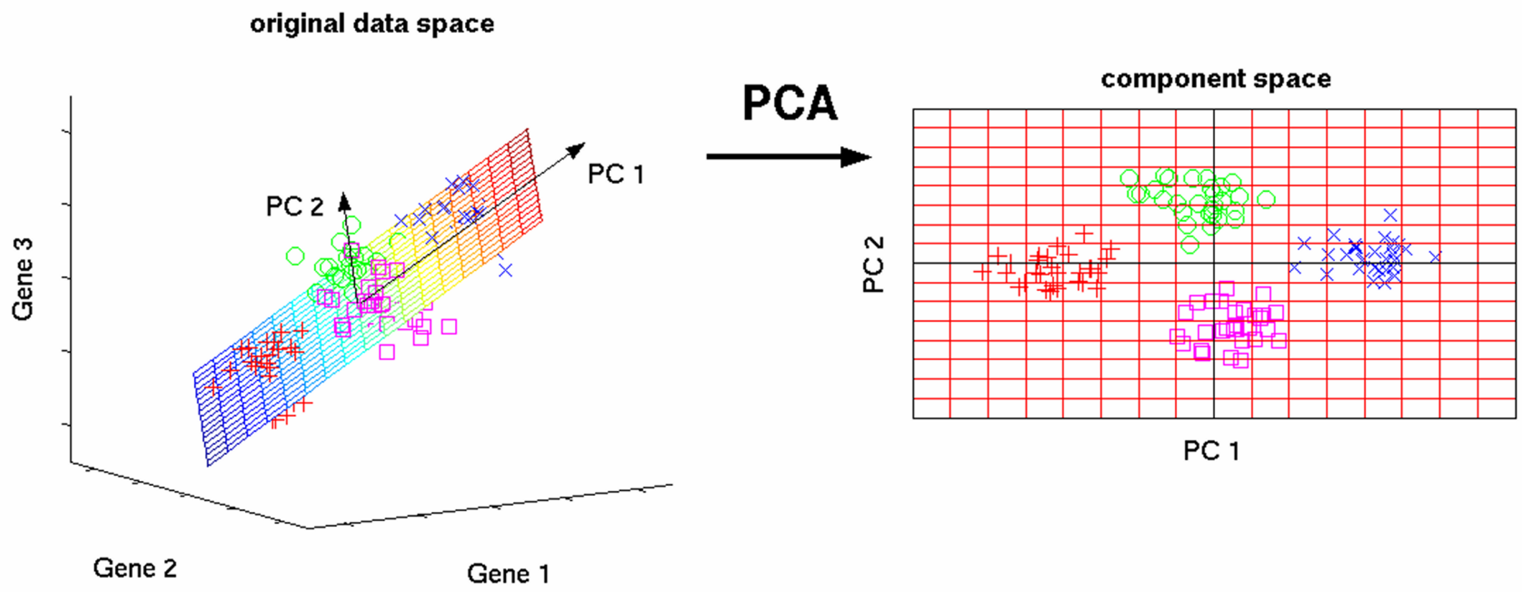
Aside Question: how do we know we have exactly the same image slice?

Mutual Information

$$MI(A,B) = -\sum_{x_1 \in A, x_2 \in B} p(x_1,x_2) \log\left(\frac{p(x_1,x_2)}{p(x_1)p(x_2)}\right)$$

- Defined as: MI(A, B) = H(A) + H(B) – H(A, B)
- Measures the dependence of the two distributions
- In image registration will be maximized when the images are aligned
- In feature selection choose the features that minimize I(A,B) to ensure they are not related

Principal component analysis



概率模型与聚类（快速过）

- Gaussian 分布： 由 mean 和 variance 决定
- MLE：最大化似然函数求参数
- GMM + EM 算法：

  E-step — 计算 responsibilities（软分配）

  M-step — 更新参数
- K-Means vs GMM：

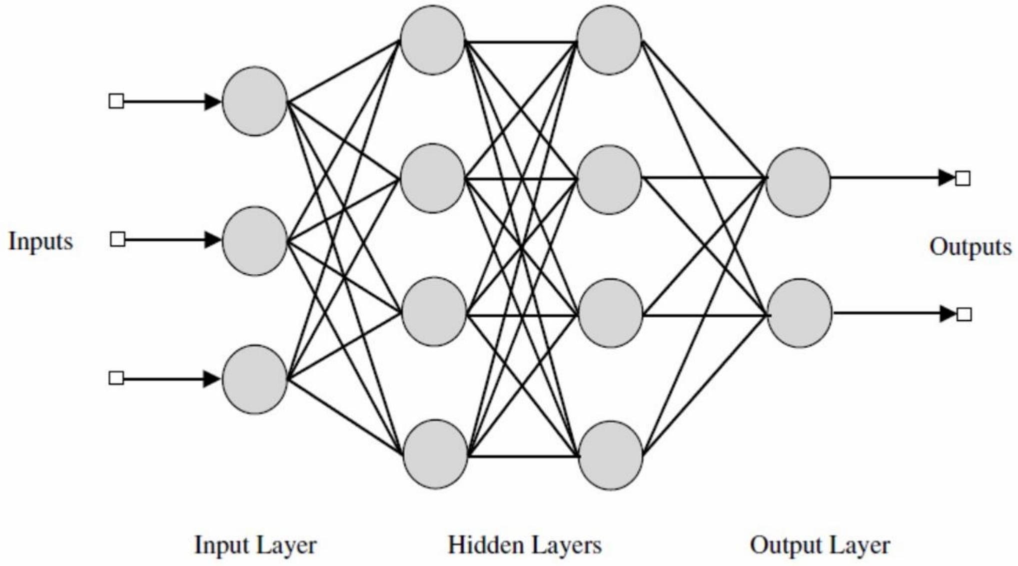
  K-Means = 硬分配 GMM = 软分配（概率）

分类方法速览

- Bayes Rule: posterior ∝ likelihood × prior
- Loss Functions: Quadratic / 0-1 / Hinge / Logistic
- Logistic Regression: sigmoid 映射到概率
- SVM:
  - Maximum margin 超平面
  - Slack variables 处理不可分
  - Kernel trick: 映射到高维空间

- 核函数: Linear / Polynomial / Gaussian(RBF)

Feed-forward neural networks



After Bishop, Pattern Recognition & Machine Learning, Chapter 5

第六部分：Machine Learning

Neural Networks 核心概念

参数计算公式：权重 = Σ(n_in × n_out)　偏置 = Σ(n_out)

✗ 参数可以用解析解求 → 不能！需要梯度下降+反向传播

✓ 2个输出单元 → 可以做 binary classification（sigmoid）

✓ Universal Approximation Theorem:单隐层理论上够用 BUT 更深的网络泛化更好

✓ Backprop: 前向→误差δ→反向传播δ→更新权重

⚠ Overfitting: 参数过多容易过拟合

模型评估

- Precision = TP/(TP+FP) — "预测为正"中真正为正的比例
- Recall = TP/(TP+FN) — "真实为正"中被正确识别的比例

⚠ 不要搞混！Precision看"预测正"，Recall看"真实正"

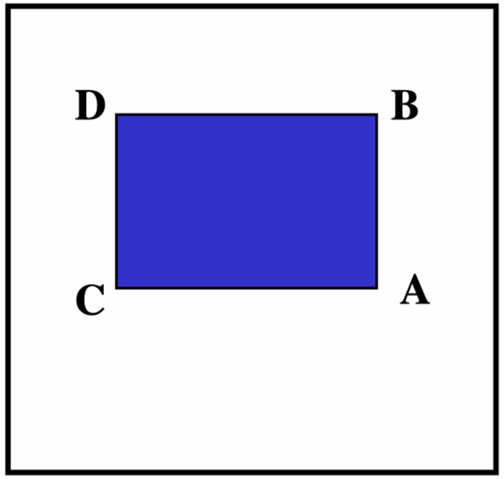
- Entropy: $H(X) = -\Sigma\, p(x) \log p(x)$
- KL Divergence: 衡量两个概率分布的差异（非对称）

Computing the sum within a rectangle

- Let A, B, C, D be the values of the integral image at the corners of a rectangle

- The sum of the original image values within the rectangle can be computed as: sum = A – B - C + D

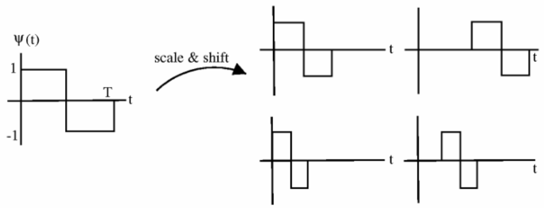
- Only 3 additions are required for any size of rectangle

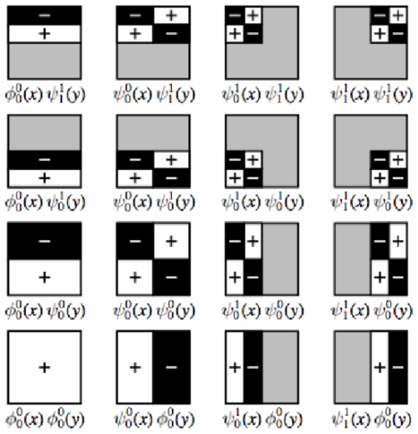


Haar wavelets

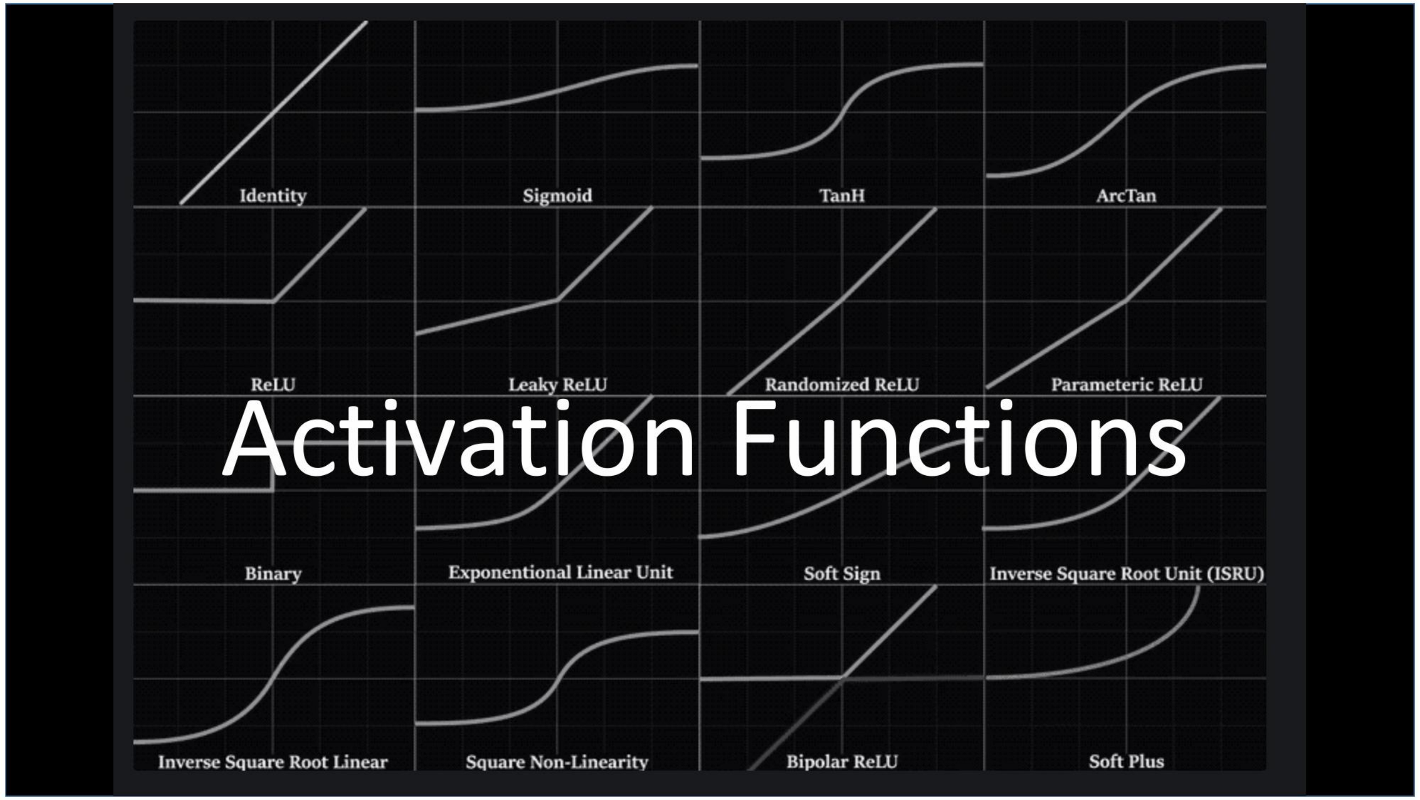
- Haar wavelets for a asset of basis functions that are very efficient to compute

- Similar to Gabor wavelets they form a complete basis





Box filters used by Viola Jones


Identity
Sigmoid
TanH
ArcTan
ReLU
Leaky ReLU
Randomized ReLU
Parameteric ReLU
Activation Functions
Binary
Exponentional Linear Unit
Soft Sign
Inverse Square Root Unit (ISRU)
Inverse Square Root Linear
Square Non-Linearity
Bipolar ReLU
Soft Plus

第七部分：Detection & DL

经典网络架构（了解）

- LeNet (1989) — 手写数字
- AlexNet (2012) — ImageNet冠军，开启DL时代
- VGGNet (2014) — 证明深度重要
- FCN — 全卷积网络用于语义分割

HOG (Histogram of Oriented Gradients)

用于行人检测 + 线性SVM

规则网格上计算梯度方向直方图

局部对比度归一化至关重要

HOG > Haar wavelets

第八部分：模拟考题 + 总结

关于聚类算法，以下哪些描述是正确的？

A. K-Means 需要预先指定K，而 Mean Shift 不需要

B. K-Means 的结果不受初始化影响，因为保证收敛到全局最优

C. Mean Shift 通过沿密度梯度方向移动找到聚集中心

D. K-Means 最小化的目标函数是簇内方差之和 (WCSS)

全连接网络：4输入→6隐藏(ReLU)→3输出(Softmax)

A. 共有 $(4\times6+6)+(6\times3+3) = 51$ 个可训练参数

B. 替换ReLU为线性激活，表达能力不受影响

C. 反向传播通过链式法则计算梯度

D. Softmax 输出3个值之和为1

第八部分：模拟考题＋总结

关于图像去噪方法，以下哪些描述是正确的？

A. 高斯滤波器是一种线性滤波器，它通过加权平均的方式平滑图像

B. 中值滤波器比高斯滤波器更擅长保留图像边缘

C. 高斯滤波器在频率域等价于一个低通滤波器

D. 增大高斯滤波器的 σ 值会使图像边缘更加清晰

关于 Otsu 自动阈值法，以下哪些描述是错误的？

A. Otsu 方法通过最大化类间方差来寻找最优阈值

B. Otsu 方法假设图像直方图具有双峰分布

C. Otsu 方法能够很好地处理光照不均匀的图像

D. Otsu 方法产生的是全局阈值，对图像中所有像素使用同一阈值

关于 SIFT 特征描述子，以下哪些描述是错误的？

A. SIFT 通过在 Difference-of-Gaussian (DoG) 金字塔中寻找极值点来检测关键点

B. SIFT 描述子最终的维度是 64 维（4×4 子区域 × 4-bin 方向直方图）

C. SIFT 通过为每个关键点分配主方向来实现旋转不变性

D. 在关键点定位阶段，SIFT 使用 Hessian 矩阵的特征值比来剔除边缘响应点

第八部分：模拟考题＋总结

关于 Viola-Jones 人脸检测器中的 AdaBoost 算法，以下哪些描述是正确的？

A. AdaBoost 从大量弱分类器中选择并组合成一个强分类器

B. 在每一轮 boosting 中，被正确分类的样本会获得更高的权重

C. 级联分类器的早期阶段使用少量特征，后期阶段使用更多特征

D. Viola-Jones 中的弱分类器基于 Haar-like 特征构建

关于模型评估指标，以下哪些描述是正确的？

A. Dice 系数和 Jaccard 指数是单调相关的，它们捕获的是相同方面的信息

B. Hausdorff Distance 计算的是两个轮廓之间所有点对距离的平均值

C. Precision 衡量的是所有正样本中被正确识别的比例

D. 当 Dice = 0.8 时，对应的 Jaccard 值小于 0.8

高频易错点 TOP 10

1. K-Means 只能处理凸/球形簇
2. Dice 和 Jaccard 单调相关，不需同时计算
3. Hausdorff 是 max of min，不是 mean
4. PCA 最大特征值 = 最大方差方向
5. Viola-Jones 不是 pose independent
6. 级联分类器早期用少量特征
7. Mutual Information 适合跨模态配准
8. 神经网络不能用解析解求参数
9. SLIC 不是机器学习方法
10. 积分图 O(1) 算矩形区域和 (A-B-C+D)

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